

Prepare a short write-up (2 pages max.) describing your processing techniques and choices above.

1. For the given obfuscated dataset comprising 2000 rows $\times$ 552 columns, we first load the dataset into a Pandas data frame to prepare for analysis.

2. It is observed that the dataset comprises various columns of different data types such as string, float, and integer.

3. To further examine the data for erroneous features, some preprocessing is performed on the data frame to check if it consists of null values. As a result, a total of 100 rows were found to contain NaN, NA, or null values therefore they are dropped from the data frame.

- A cleaned version of the data frame is then prepared which consists of columns containing only float or integer values to perform exploratory data analysis using mathematical techniques (as said techniques cannot be applied on string values, they are removed from the cleaned dataset).

- The string values are considered best to be dropped out of the cleaned dataset, as only a few columns containing string values can be categorized as ordinal data, however, there is no clear scale for conversion of those ordinal values into integer values, hence, they are dropped.

- Now, the cleaned dataset comprises 1900 rows $\times$ 549 columns for analysis.

4. On the cleaned dataset, further preprocessing is performed to find and eliminate outliers. This is done in order to make the dataset less obfuscated and allow the statistical analysis to be conducted properly without shifting the mean values toward the outliers.

- The method used for detecting outliers is IQR (Inter Quartile Range) which is used to statistically sort the data from first quartile (Q1) $\rightarrow$ Median $\rightarrow$ third quartile (Q3).

- Generally, IQR uses 1.5 as its scale to detect outliers because it most closely follows Gaussian distribution, however, for our dataset, we have extended the range by increasing the 1.5 multiplier to 6.

- This is done in order to allow the dataset to actually filter out values that are extremely beyond the given range which can be defined as outliers with full confidence. As a result, an average of 7 outliers are detected in each column.

- Now a filtered dataset out of the cleaned dataset is formed which consists of only those rows from each column that lie between the range of lower bound quartile and upper bound quartile, and void of outliers.

- The filtered dataset comprises 70 rows $\times$ 549 columns for further analysis.

5. Now, we identify out of 549 columns, which pair of columns are highly correlated to understand which features are necessary to keep in the dataset.

- A correlation coefficient (Pearson's correlation coefficient) is a statistical measure that indicates the strength and direction of a linear relationship between two variables/features.

- To find highly correlated columns, the correlation matrix is used for the entire filtered dataset where using permutation and combination for each feature/column, another feature/column is computed that has the maximum correlation of the first feature/column.

- Upon computing the correlation for a pair of features/columns, we observe a range of correlation values ranging from -1.0 to +1.0, where -1.0 indicates high negative correlation, 0 indicates zero or no correlation, i.e. no linear relationship exists between the two features, and +1.0 indicates high positive correlation.

- A threshold is used to filter out only those pairs of features/columns that are either highly positively correlated or highly negatively correlated as they will have a clear positive or negative linear relationship between the two corresponding features.

- Therefore a high threshold cutoff at 0.8 for positive correlation and -0.8 for negative correlation yields about 48042 rows of pairs of features observed from all permutations and combinations of two features that have high correlation amongst them.

6. As we do not require all pairs of highly correlated features, we filter out only those pairs where either feature corresponds to the target variable and is highly correlated to it.

- Here, we define threshold cutoff at 0.5 for positive correlation and -0.5 for negative correlation as there exists no pair such that it has an extremely high correlation above 0.8 and below -0.8 with the target variable - 'experimental_property'.

- This yields about 153 rows of pairs of features that are observed to have a high correlation with the target variable.

7. Now, we use Principal Component Analysis which is an unsupervised machine learning technique used to reduce the dimensionality of the dataset and find the principal components that contribute to the most information in a large dataset.

- Using PCA we plot 3 different 2D scatter plots to differentiate between the obfuscated dataset with outliers, the filtered dataset without outliers but taking all pairs of correlated features and the filtered dataset without outliers comprising only the highly correlated features with the target variable.

- 7.1 As observed from the first plot, outliers create a huge clump of all scatter points that provide no information upon observation.

- 7.2 As observed from the second plot, the removal of outliers improves the pattern observation in the scatter plot however there is no clear relationship with the target property.

- 7.3 As observed from the third plot, we use only the highly correlated columns as observed from the second correlation matrix with respect to the target property. This improves the pattern generation in the scatter plot and its relationship to the target variable. From further observation, it can be concluded that the data points on the left side of the graph correlate to higher values of the target variable and that on the right side of the graph correlate to lower values.

Bonus Question.: Compare the resulting plot with PCA.

1. For comparison of PCA with other non-linear dimensionality reduction methods, I have used two separate methods:

1.1 t-SNE:

- It stands for t-distributed stochastic neighbor embedding and is a statistical method for visualizing high-dimensional data by giving each data point a location in a two or three-dimensional map. First, t-SNE constructs a probability distribution over pairs of high-dimensional objects such that, similar objects are assigned a higher probability while dissimilar points are assigned a lower probability. Second, t-SNE defines a similar probability distribution over the points in the low-dimensional map, and it minimizes divergence between the two distributions with respect to the locations of the points in the map.
- As observed from the scatter plot of t-SNE, in comparison to the PCA scatter plot, the data points seem to have shifted in a manner that the previously left side corresponding to higher values in target property, now can be seen towards the right side corresponding to their higher probability of similarity with the target variable and closer in intra-cluster distance.
- Similarly, the previously observed right side in PCA, corresponding to the lower values in the target property, now can be seen on the left side with lesser similarity, and more distance between the points inside the cluster.

1.2 UMAP:

- It stands for Uniform Manifold Approximation and Projection and is a dimensionality reduction technique that helps visualize high-dimensional data in a lower two or three-dimensional space. It has a dimension reduction technique that can be used for visualization similarly to t-SNE, but also for general non-linear dimension reduction.
- As observed from the scatter plot of UMAP, in comparison to the PCA scatter plot, the data points are seen in a clear pattern with a direct relationship to the target property. As previously observed, the data points on the left side of the graph correspond to higher values of target property as well as they are closer in intra-cluster distances.
- Similarly, as previously observed, the data points on the right side of the scatter plot correspond to the lower values of the target property and have higher intra-cluster distances.

Bonus Question.: Uncover hidden patterns and find non-trivial relationships in the data.

2. A hidden pattern as observed in the PCA scatter plot is that there seems to be some degree of linear separability between groups of different target properties, particularly between the higher values (3 and 4) and lower values (-1, 0, 1). This could imply that the features that contribute to these principal components might be good candidates for predicting the target property in a linear classification model. If Principal Component 1 is

fixed at 0 while Principal Component 2 varies, the pattern observed to the left of Principal Component 1 = 0 is the mirror image of the pattern observed to the right of Principal Component 1 = 0. The layout of points may suggest potential correlations between the principal components and the target property. For instance, an increase in both principal components might be associated with an increase in the target property value, particularly noticeable in the transition from the lower left to the upper right of the plot.

3. Hence, some non-trivial relationships in the dataset could be obtained through this pattern are:
 - a. Specific features in the dataset that are heavily influenced by Principal Components 1 and 2 may be significantly correlated with the target property. For instance, if the data includes features like income and education level, and these are major contributors to the principal components, there may be a non-trivial relationship between these socioeconomic factors and the target property.
 - b. There appears to be a gradient change in the target property values as we move from the middle to the upper-left and upper-right corners of the plot. This suggests that the target property might increase with an increase in the value of Principal Component 2.
 - c. There may be threshold effects at certain levels of the principal components beyond which the target property jumps to a higher or lower value sharply. This could be visible in how the clustering of colors changes abruptly in certain areas of the plot.
 - d. The density of points in certain areas might indicate common or typical states of the underlying features, whereas sparse areas could indicate unusual or rare combinations of features.